LANGSMITH · MANAGED DEEP AGENTS

LangSmith Sandboxes

What a sandbox is, and how a Managed Deep Agent uses one

Notebooks, agent source and docs: [repo link]

SANDBOXES

Agents that write code need a computer of their own.

```
agent › execute
pip install pandas
python3 analysis.py
rm -rf ./data # a bad guess
curl evil.example # injected
```



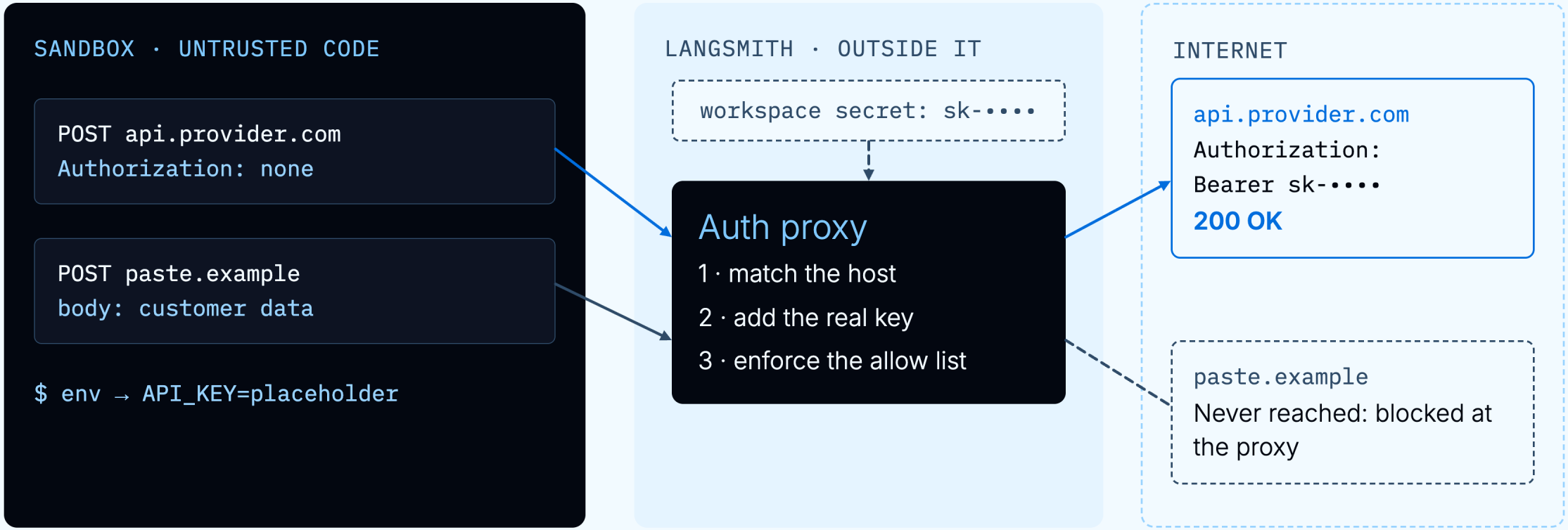
A LangSmith Sandbox

- A Firecracker microVM with its own kernel
- Ready in under a second
- Root access and its own filesystem
- Up to 16 vCPU and 64 GiB memory

Wrong or prompt-injected, the damage stays inside the sandbox.

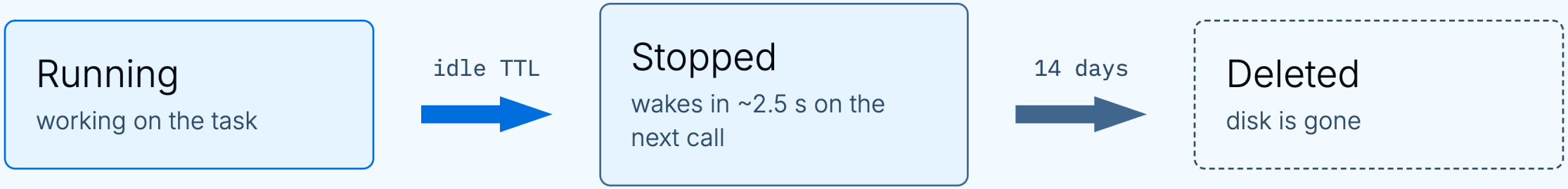
THE AUTH PROXY

Secrets stay outside.



LIFECYCLE

A sandbox is scratch space with a lifecycle.



SURVIVES A STOP

Files: the data, code and outputs from earlier turns

DOESN'T

Running processes: it boots fresh, so a live app stops

DEMO 1 · NOTEBOOKS/01_SANDBOX_BASICS.IPYNB

Create a sandbox, run code, and try to break out.

- ready in <1s
- files in, chart out
- no key inside, call still works
- example.com blocked
- stop, wake, files kept
- boot from a snapshot

MANAGED DEEP AGENTS

An entire runtime around your agent.

YOU PROVIDE

Business logic

instructions
tools
skills
model



Deep Agents harness agent loop · planning · filesystem · subagents

MANAGED RUNTIME ON LANGSMITH

Agent Server
threads, durable runs

Sandboxes
code execution

LangSmith Tools
governed tool access

LLM Gateway
models, spend limits

Context Hub
prompts, skills, memory

Tracing
every model and tool call

ANATOMY

An agent is a folder.

Managed context

instructions.md · skills/

Your code

agent.py · tools/ · middleware/

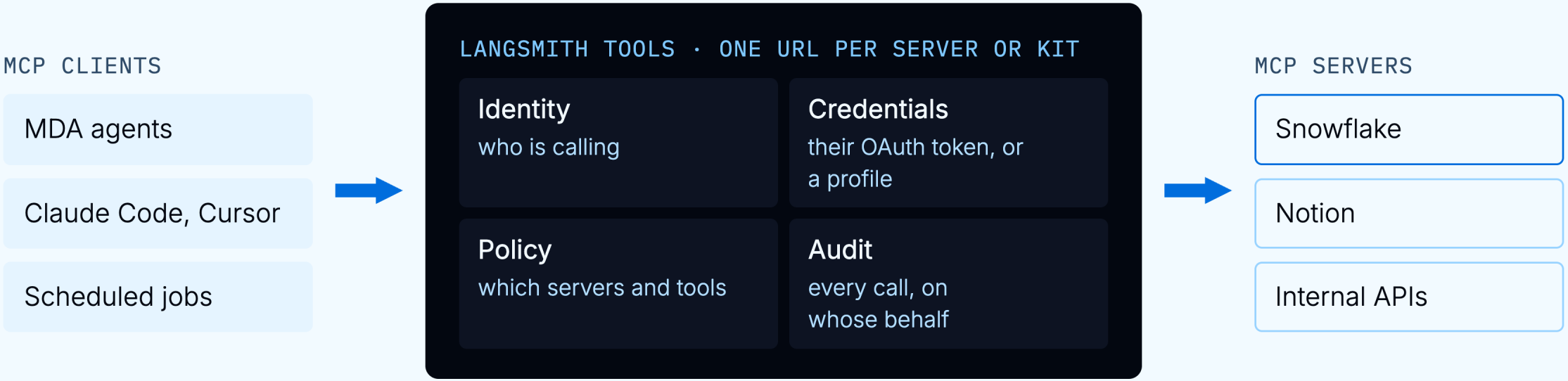
Managed config

sandbox/ · identity.py · .env

```
snowflake-analyst/
├─ agent.py                  # model, tools, middleware
├─ instructions.md           # system prompt
├─ skills/dashboard/         # loaded on demand
├─ tools/mcp.py              # Snowflake, via LangSmith Tools
├─ tools/artifacts.py        # share charts and dashboards
├─ middleware/               # keeps links out of the model
├─ sandbox/__init__.py       # one sandbox per thread
├─ sandbox/setup.sh          # baked into a snapshot
├─ identity.py               # who may call it
└─ .env                      # becomes deployment secrets
```

LANGSMITH TOOLS · THE TOOL GATEWAY

One gateway for every agent's tools.



WITHOUT A GATEWAY

Every agent carries its own server URLs, keys and tool lists. Changes mean redeploys.

WITH LANGSMITH TOOLS

Register once, connect once, govern centrally. Agent code never handles a credential.

DEMO 2 · THE DEPLOYED AGENT IN LANGSMITH STUDIO

TURN 1

"Build me a revenue dashboard: monthly revenue trend, revenue by region by year, and the top-line KPIs."

TURN 2 · SAME THREAD

"Add a 6-month revenue forecast to the dashboard."

Query Snowflake as you → Write data and code → Run it in the sandbox → Share chart and link

---

0 Snowflake calls in turn 2. The data is still in the sandbox, so the agent backtests a model on the last 6 known months, fits it, and ships dashboard v2.

SHIPPING IT

One command: mda deploy

```
$ mda dev # local server + Studio
$ mda deploy --agent-environment development
✓ bake sandbox/setup.sh into a snapshot
✓ sync instructions.md + skills/ to Context Hub
✓ forward .env to deployment secrets
✓ build and deploy a new revision
status: DEPLOYED
```

What you don't build

- ✓ Checkpointer and store
- ✓ Per-thread sandboxes
- ✓ OAuth and token storage
- ✓ Slack endpoints, crons
- ✓ Traces for every call

development › staging › production

Serverless: scales to zero

Dedicated: always on

GitHub: preview build per PR

NEXT STEPS

Try it in your own workspace.

01 · 15 minutes

Run the sandbox notebook

Only a LangSmith API key needed.

02 · an afternoon

Register one governed tool

An MCP server in LangSmith Tools that people use as themselves.

03 · a sprint

Build one agent on MDA

Pick a use case that needs code: analysis, reports, files.

THE REPO · [repo link]

notebooks/ · mda-snowflake-analyst/ · docs/snowflake-oauth-setup.md · docs/mda-workflow.md

Sandboxes: generally available · Managed Deep Agents and LangSmith Tools: beta, US cloud